

# One definition. Every team. *The same answer.*

Governed conversational analytics on Snowflake semantic views, with a test that proves the answers stay identical.

| TEAM NAME     | TEAM LEADERS                  | TEAM SIZE | PROBLEM STATEMENT                                                                                                                                                       |
|---------------|-------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3M-ONTOLOGIST | Mahantesh · Devansh · Sathyam | 3         | Teams compute the “same” supply chain metric differently, get different numbers, and cannot prove which is right. The wrong one quietly clears non-compliant suppliers. |

## 1 · PROBLEM BRIEF

Two numbers.  
Same 420,000  
*receipt lines.*

*“Same question, two answers. The hour gets spent on whose number is right, not on what to do.”*

LEGACY

AVERAGE OF MONTHLY AVERAGES

0.882631

Supplier on-time delivery, as most dashboards compute it today.

Δ 0.006807 — enough to change who passes

GOVERNED

EVERY RECEIPT LINE COUNTS ONCE

0.875824

Defined once as a Snowflake semantic view, tested against atomic facts.

Source SEMANTIC.SC\_SUPPLIER\_LEGACY\_DEFECT vs SC\_SUPPLIER

## 1 · PROBLEM BRIEF

# A wrong pass makes *no noise*.

### BUSINESS PROBLEM

One metric, a different number in every team's tool. The disagreement is found in a review meeting, not in a test.

### TARGET USERS

Planning

Procurement

Logistics

Logistics · EU

Ontology steward

### PAIN → FIX

~~Hours arguing over whose number is right.~~ Metrics defined once as semantic views and proven identical for every persona, every day.

### INDUSTRY

Manufacturing supply chain. ERP, WMS, TMS, IoT and supplier contracts, each with its own idea of “on-time”.

12 / 300

suppliers wrongly cleared by the legacy definition

+22%

bigger “compliant” list than the truth supports

0

suppliers wrongly escalated, so nobody complains

*A false fail is disputed within a week. A false pass is never rechecked.*

# It *tests its own agreement*, every day.

And keeps a deliberately broken metric deployed, so you can watch the test catch it.

|                            | TYPICAL BI DASHBOARD | TYPICAL TEXT-TO-SQL CHAT | SUPPLY CHAIN ONTOLOGY                     |
|----------------------------|----------------------|--------------------------|-------------------------------------------|
| Where a metric is defined  | In each report       | In each prompt           | Once, in a governed registry              |
| Same number for every team | Hoped for            | Not guaranteed           | Proven daily, zero spread                 |
| Access control             | App layer            | App layer                | Snowflake role and row policy             |
| Numbers in the answer      | Not applicable       | Can be invented          | Invented figures discarded                |
| Contracts and free text    | Left out             | Left out                 | Read by AI_EXTRACT, 2,700 of 2,700 fields |
| Cost of disagreement       | Unknown              | Unknown                  | 12 suppliers · \$178,832 hidden           |

# Plain English in. *A governed number out.*

The model never writes SQL. It picks registered metrics; anything unregistered cannot be asked.

Follow-ups work. “And by supplier region?” is understood, and every turn is re-checked against the persona’s grants.

Invented figures are discarded. Prose with a number the rows lack is replaced by a template.

Speak it. Whisper runs in the browser. Audio never leaves the device.

1.7 s

90.0%

repeat question, was 16–20 s

on a scored 60-question eval

sc-ontology-app.vercel.app/ask

Period · Trailing 12 months · 2025-09-01 to 2026-08-31 (12 complete months)

What is our supplier on-time delivery?  
as Ontology Steward · Trailing 12 months

GOVERNED ANSWER

The question asks for an overall supplier on-time delivery figure, which is the inbound metric measuring receipt lines received on or before the supplier’s promised date.

1 row

executed as SC\_ONTOLGY\_STEWARD (secondary roles disabled, row scope ALL REGIONS) · 

certified · 13/13 quality checks · 15/15 drift-free

 · Trailing 12 months

Supplier On-Time Delivery for the trailing 12 months stands at 87.17%. The target for this metric is 90.00%, which is not met. This metric reflects REALIZED scope data from the governed registry.

Supplier On-Time Delivery

drift PASS

87.17%

target 90.00% · below target

Share of purchase-order receipt lines received on or before the date the supplier promised. Weighted by receipt line, so a supplier with many lines influences the result in proportion to its volume.

Show the rows behind this

● Governed Forward Outlook (as\_of\_scope: PREDICTED)

TARGET\_BREACH

Accuracy on held-out data: 99.3%

Interactive simulation ›

|                                                                                                   |                                                                                                       |                                                                                                     |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <div>Films</div> <div>Breach probability</div> <div>87.42%</div> <div>100.0% (UNREACHABLE)</div>  | <div>Abrasives</div> <div>Breach probability</div> <div>86.94%</div> <div>100.0% (UNREACHABLE)</div>  | <div>Adhesives</div> <div>Breach probability</div> <div>87.24%</div> <div>99.9% (UNREACHABLE)</div> |
| <div>Safety</div> <div>Breach probability</div> <div>87.05%</div> <div>99.9% (UNREACHABLE)</div>  | <div>Filtration</div> <div>Breach probability</div> <div>87.05%</div> <div>99.9% (UNREACHABLE)</div>  | <div>Tapes</div> <div>Breach probability</div> <div>87.17%</div> <div>99.8% (UNREACHABLE)</div>     |
| <div>Medical</div> <div>Breach probability</div> <div>87.35%</div> <div>99.7% (UNREACHABLE)</div> | <div>Electronics</div> <div>Breach probability</div> <div>87.48%</div> <div>99.0% (UNREACHABLE)</div> |                                                                                                     |

Target is 5.12 standard deviations of ordinary month-to-month variation BEYOND current capability for Films (mean 0.8742, sd 0.005 over 22 full months). The z-score and verdict are the reliable outputs; the probability is a logistic approximation to the normal CDF and 22 monthly observations cannot support precise tail probabilities.

sc-ontology-app.vercel.app/consistency

GOVERNED ANALYTICS WORKSPACE

Cross-Persona Consistency

Compare governed metrics across real Snowflake roles and row scopes.

PERSONAS

4

Real roles, enforced by Snowflake RBAC

SHARED METRICS

15

Served by more than one semantic view

CURRENT AGREEMENT

EXACT

Zero spread across all bindings

DIVERGENCE BEFORE FIX

0.68%

Supplier OTD, same question, two answers

Live cross-persona execution

Pick a governed metric. The app re-executes it once per persona role, through each semantic view that persona can reach, using that role's own grants. Identical values mean the definition — not the team — decides the answer.

GOVERNED METRIC

Supplier On-Time Delivery

Run as all personas

Share of purchase-order receipt lines received on or before the date the supplier promised. Weighted by receipt line, so a supplier with many lines influences the result in proportion to its volume.

Persona access model

| Persona                    | Snowflake role      | Focus                                                                                                            | Row scope        | Semantic views                                                                                                                     |
|----------------------------|---------------------|------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Ontology Steward           | SC_ONTOLOGY_STEWARD | Owns the semantic layer. The only persona that sees every view, the drift control and the consistency evidence.  | ALL REGIONS      | SC_CONTRACT, SC_DEMAND, SC_FULFILLMENT, SC_INVENTORY, SC_LANDED_COST, SC_MANUFACTURING, SC_ONTOLOGY_360, SC_SUPPLIER, SC_TELEMETRY |
| Procurement Analyst        | SC_PROCUREMENT      | Supplier delivery, fill and purchase price variance on the inbound side.                                         | ALL REGIONS      | SC_CONTRACT, SC_ONTOLOGY_360, SC_SUPPLIER                                                                                          |
| Logistics Manager          | SC_LOGISTICS        | Outbound delivery performance and cost to serve, all regions.                                                    | ALL REGIONS      | SC_FULFILLMENT, SC_LANDED_COST, SC_ONTOLOGY_360, SC_TELEMETRY                                                                      |
| Logistics Manager, EU      | SC_LOGISTICS_EU     | The same accountability as SC_LOGISTICS, restricted to EU destinations. Same metric definitions, different rows. | SHIP_REGION = EU | SC_FULFILLMENT, SC_LANDED_COST, SC_ONTOLOGY_360, SC_TELEMETRY                                                                      |
| Demand & Inventory Planner | SC_PLANNER          | Inventory cover, stock value and demand plan accuracy.                                                           | ALL REGIONS      | SC_DEMAND, SC_FULFILLMENT, SC_INVENTORY, SC_ONTOLOGY_360                                                                           |

Row scope and metric definition are deliberately separate concerns. The EU-scoped logistics role sees fewer rows because of a row access policy,

FEATURE 02 · CONSISTENCY

One metric. Four real roles. *The same number.*

Executed as each Snowflake role, with secondary roles off, so the database enforces access.

Checked against atomic SQL. Two views cannot agree while both being wrong. Runs daily at 06:00 UTC.

A broken control stays live. 0.882631 against the correct 0.875824 proves the test can fail.

15

/

15

4

bindings match, zero spread

personas, same answer

FEATURE 03 · ONTOLOGY

The diagram *cannot drift* from what is queried.

Read live from INFORMATION\_SCHEMA. Entities and relationships come from the deployed view.

Hierarchies are measured. Every rollup is tested as a true functional dependency. Two plausible ones were rejected.

Portable. OWL/SKOS export as Turtle or JSON-LD, with SCOR codes as exactMatch and closeMatch.

12

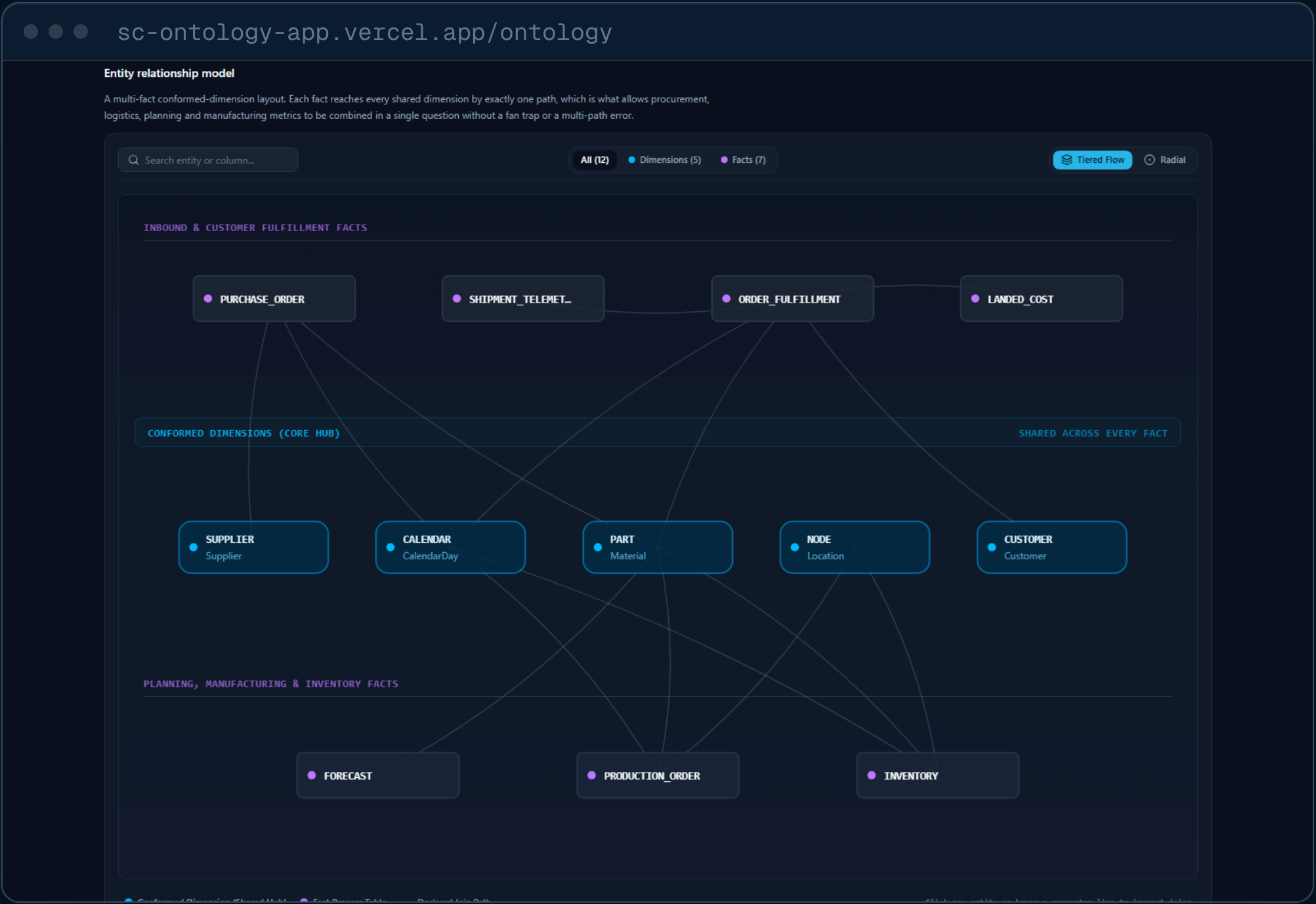
entities

15

relationships

10

rollups proven



sc-ontology-app.vercel.app/impact

Impact

What governed definitions are worth, measured: consistency, accuracy, decisions, money and time, each with its source.

Every figure below is read from a governed object at page load and carries its source. Exactly one input is an assumption: how long a disputed cross-team metric takes to settle by hand. It is labelled, and the one row derived from it says so.

| Pillar      | Measure                                                                           | Value                                                                  | Basis      | Source                               |
|-------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------|------------|--------------------------------------|
| Consistency | Metric bindings resolving to the canonical value                                  | 15 / 15                                                                | MEASURED   | METRIC_DRIFT_RESULT (run 2026-09-29) |
| Consistency | Largest spread between personas on the same metric                                | 0                                                                      | MEASURED   | METRIC_DRIFT_RESULT                  |
| Accuracy    | Conversational eval pass rate (golden questions, run as each persona)             | 90% (54 / 60)                                                          | MEASURED   | AGENT_EVAL_RUN (claude-sonnet-4-5)   |
| Accuracy    | Invented-metric questions correctly refused                                       | 7 / 8                                                                  | MEASURED   | AGENT_EVAL_RUN                       |
| Accuracy    | Cortex Agent reproduces the canonical definition (agent vs. app vs. registry SQL) | 4 / 4 reconciled, 0 diverge (4 differ only by the governed as-of rule) | MEASURED   | AGENT_PARITY_RESULT                  |
| Accuracy    | Contract terms extracted correctly by AI_EXTRACT                                  | 100.0% of fields                                                       | MEASURED   | V_CONTRACT_EXTRACTION_ACCURACY       |
| Decisions   | Suppliers the legacy definition wrongly clears                                    | 12 of 300 (compliant list inflated 22%)                                | MEASURED   | V_DIVERGENCE_IMPACT                  |
| Money       | Claimable contract penalties on governed breaches                                 | \$815,222 across 67 suppliers                                          | MEASURED   | V_CONTRACT_IMPACT                    |
| Money       | Penalties hidden by the legacy definition (breached, shown compliant)             | \$178,832 across 18 suppliers                                          | MEASURED   | V_CONTRACT_IMPACT                    |
| Governance  | Contracts measuring OTD with the non-governed definition                          | 79 of 300                                                              | MEASURED   | V_CONTRACT_IMPACT                    |
| Time        | Mean time to a governed, persona-scoped answer                                    | 18.0 s (p95 20.1 s)                                                    | MEASURED   | AGENT_EVAL_RUN                       |
| Time        | Manual reconciliation of a disputed cross-team metric (baseline)                  | 4 h                                                                    | ASSUMPTION | replace with your own figure         |
| Time        | Speed-up versus the manual baseline                                               | 799x faster                                                            | ASSUMPTION | row 11 measured / row 12 assumed     |

How the scorecard is assembled

SOL PROVENANCE >

FEATURE 04 · IMPACT

Every claim carries *its source*.

MEASURED or ASSUMPTION, labelled on every row, with the object it came from.

Contracts become money. AI\_EXTRACT reads 300 free-text contracts and holds them against the governed metric.

Definition conflicts surface. 79 contracts encode the non-governed definition of on-time.

\$815K

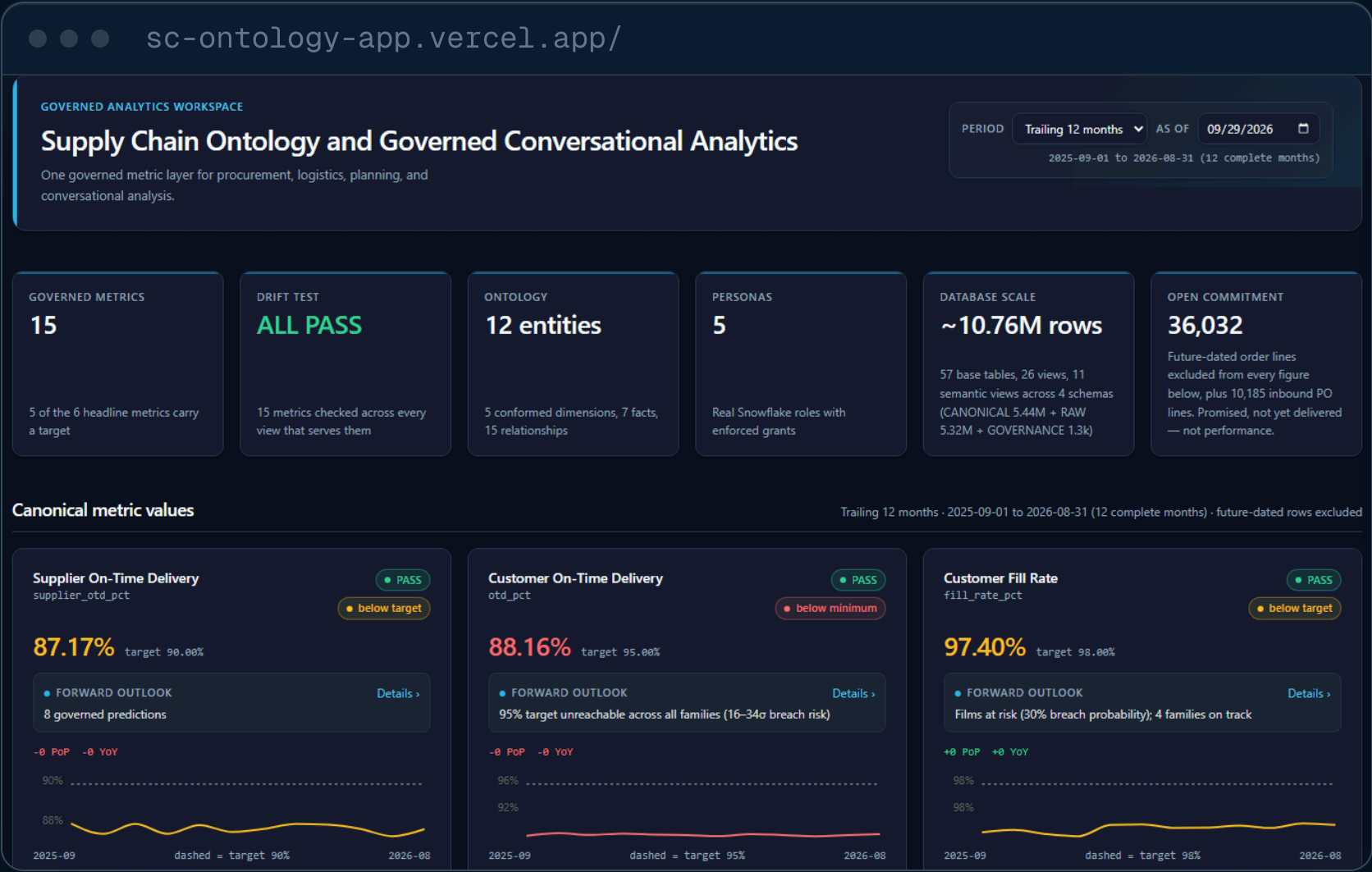
claimable penalties

\$179K

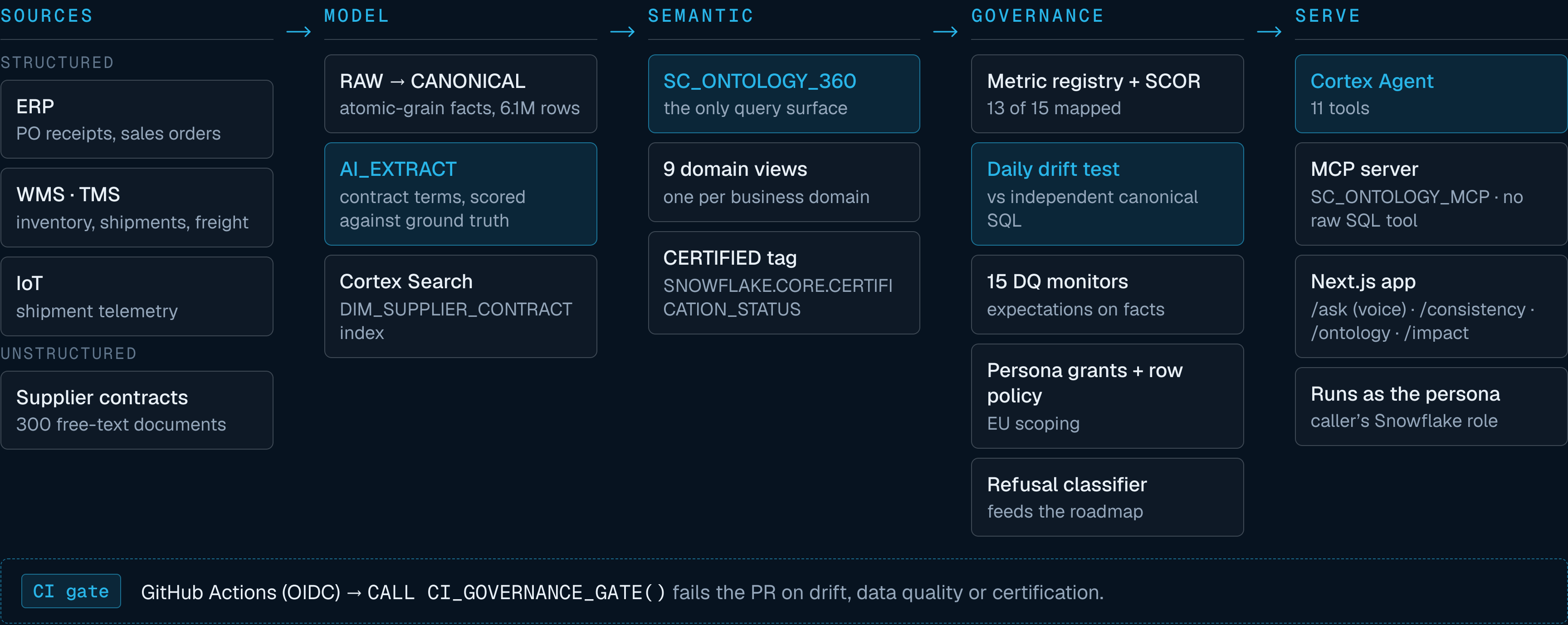
hidden by the legacy metric

# Built for operations, *not just reports.*

Plus governed forecasts shown only with their backtested accuracy, a guided tour, light and dark themes, and a shareable period with an as-of date.



# Everything queries one surface: *the ontology*.



# Fourteen layers, *one skill each*.

| LAYER                          | SQL MODULE     | COCO SKILL                                        |
|--------------------------------|----------------|---------------------------------------------------|
| Sources and facts              | 00a-00d, 11    | sql-author                                        |
| Semantic views, agent          | 00e, 07, 10    | agent-studio                                      |
| Roles, row policy, grants      | 00a, 00f       | data-governance                                   |
| Drift test, schedule, alert    | 00f, 04, 06    | data-quality · snowflake-tasks · alert            |
| Forecast, breach prediction    | 07b, 08        | machine-learning                                  |
| Contract extraction            | 16             | document-intelligence · cortex-ai-function-studio |
| SCOR industry alignment        | 18             | sql-author                                        |
| Certification, DQ, trust badge | 19             | certify-object · data-quality                     |
| MCP server for other agents    | 20             | sql-author                                        |
| Refusals → ontology roadmap    | 21             | cortex-ai-function-studio                         |
| CI governance gate             | 22, workflow   | ci-cd                                             |
| Weekly breach digest           | 23             | snowflake-tasks · notification                    |
| Impact of changing a view      | –              | lineage                                           |
| App, voice, OWL export         | app/ , app.yml | snowflake-apps                                    |

## HOW MODULES PLUG TOGETHER

Three contracts. Nothing else is shared.

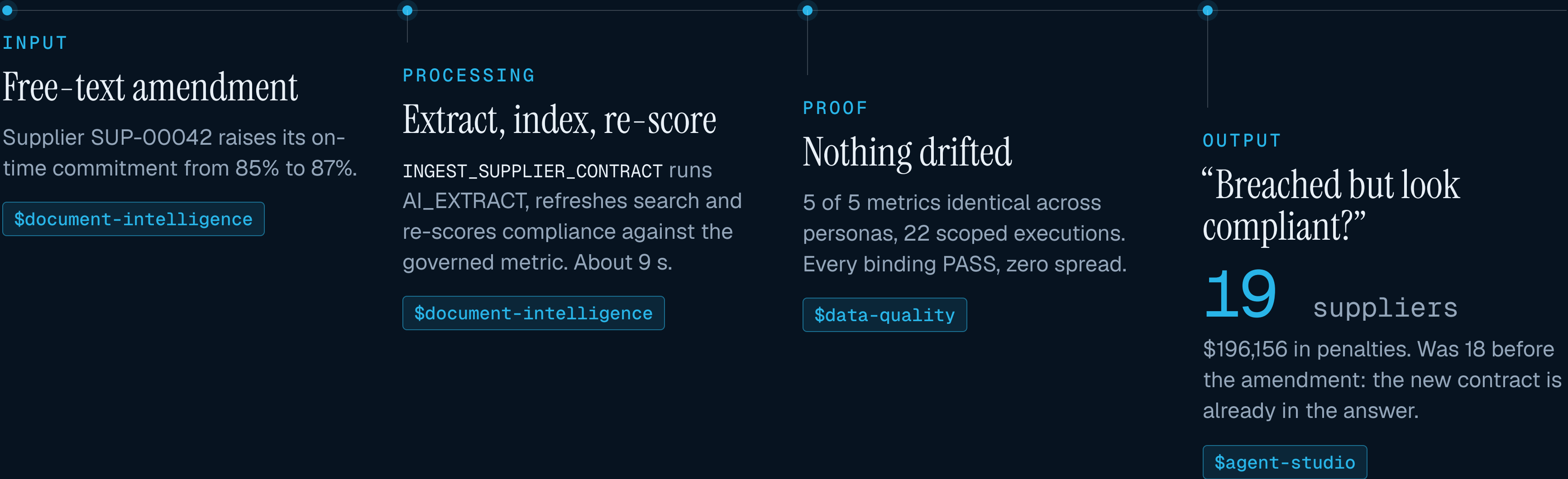
01 Atomic CANONICAL facts  
the only source of truth

02 Metric registry  
METRIC\_DEFINITION +  
METRIC\_BINDING

03 Persona access  
PERSONA\_VIEW\_ACCESS + row access  
policy

rebuild.mjs rebuilds all of it,  
6.1M rows, in about 4.5 minutes.

# A contract amendment lands. *The answer changes in nine seconds.*



\$178,832

in contract penalties the legacy metric hides, across 18 suppliers.

\$815,222

claimable across 67 suppliers

12

suppliers wrongly cleared

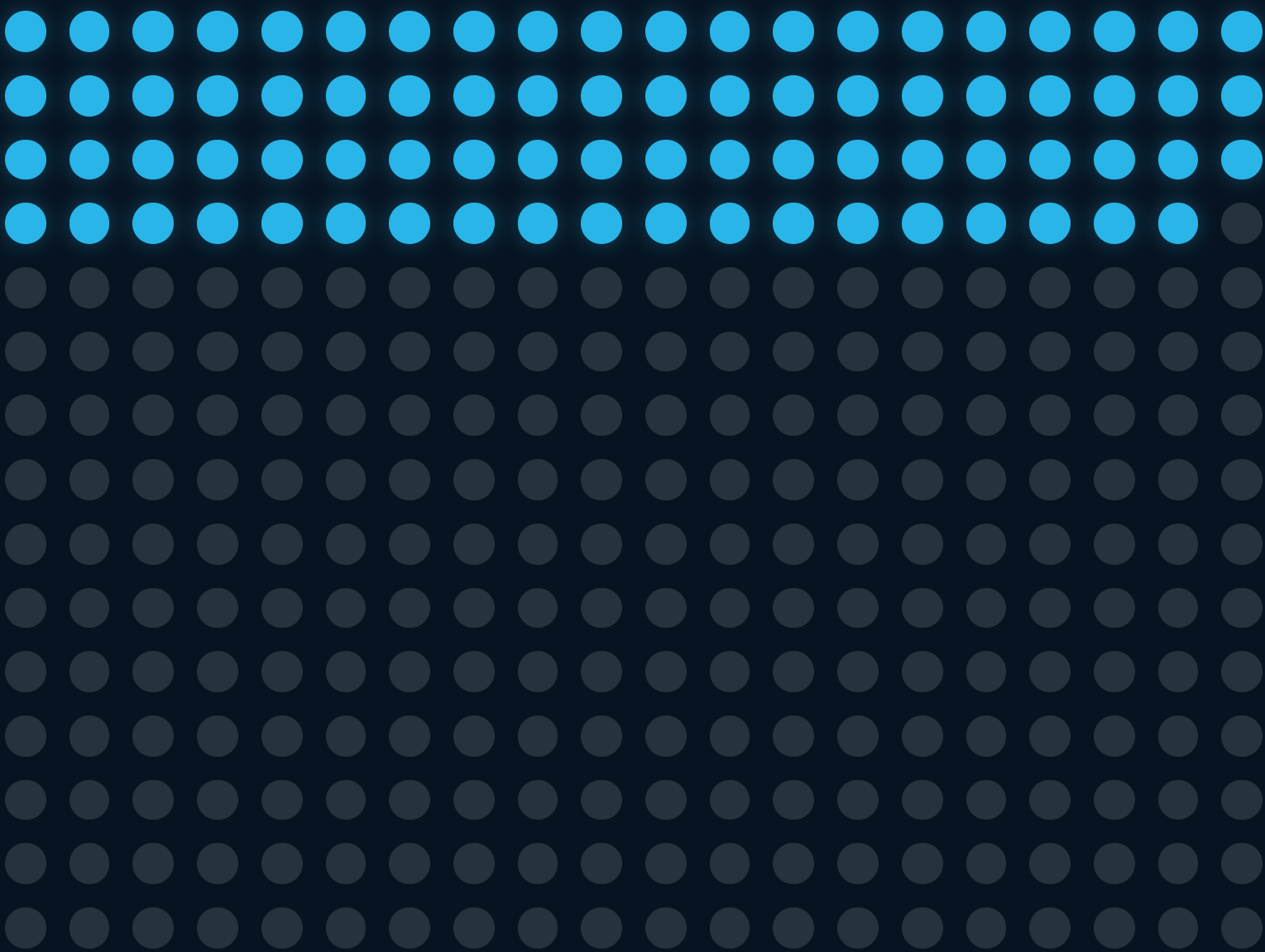
|             |                                        |                                      |
|-------------|----------------------------------------|--------------------------------------|
| CONSISTENCY | Bindings matching canonical value      | 15 / 15                              |
| CONSISTENCY | Same metric, three personas, own roles | 5 / 5                                |
| ACCURACY    | Conversational eval, 60 questions      | 90.0%                                |
| ACCURACY    | Agent vs app vs canonical SQL          | 4 / 4                                |
| ACCURACY    | Contract fields extracted correctly    | 2,700 / 2,700                        |
| TRUST       | Data-quality monitors passing          | 15 / 15                              |
| TIME SAVED  | vs 4 h manual reconciliation (assumed) | 799 <sub>x</sub>                     |
| TIME        | Governed answer, then repeat           | 18.0 <sub>s</sub> → 1.7 <sub>s</sub> |

Source GOVERNANCE.V\_IMPACT\_SCORECARD · live at /impact. Data is synthetic; targets are illustrative.

# 79 of 300 contracts *write the flaw into the agreement.*

They define on-time as a monthly average. The enforceable number and the governed number differ by construction.

*A dashboard cannot renegotiate a contract. The ontology can flag every one.*



● 79 monthly-average clause    ● 221 other

# A new metric is *one row*.

A new source is one canonical fact. IoT and contracts were both added that way, without touching the base semantic layer.

Finance close

Clinical ops

Retail

Beyond the demo: the same pattern applies anywhere two teams compute “the same” metric.

---

## NO-CODE PICKUP

Drift test, registry page, consistency runner and /ask all see a new metric. No code change.

---

## THE ROADMAP WRITES ITSELF

Refused questions are classified; unregistered-metric demand is ranked. Forecast totals and schedule adherence lead.

---

## PORTABLE ONTOLOGY

OWL/SKOS export as Turtle or JSON-LD. SCOR codes as exactMatch and closeMatch.

---

## NEW DOCUMENT TYPES

Quality certificates, rate confirmations, customs declarations. Extract, score, route low certainty to review.

---

## EVERY AGENT, SAME NUMBER

MCP exposes governed metrics, executing as the caller. EU logistics 0.8959, global 0.8855.

---

## GOVERNANCE IN THE BUILD

A PR that drifts a binding or certifies the defect view fails CI. Proven red and green.

WHY THIS IS BEST

# Proven on *all three* judging focuses.

REAL-WORLD RELEVANCE

Models the chain in the brief and prices the divergence in decisions.

12 suppliers

\$178,832 in penalties the legacy metric hides

TECHNICAL EXECUTION

Definitions are proven, not asserted, and a broken control stays deployed.

15 / 15

zero spread · 203 of 203 tests passing

SOLUTION COMPLETENESS

Structured and unstructured data, scored chat, MCP, a CI gate, OWL export.

90.0%

60-question eval · 4 of 4 parity · 2,700 fields

One definition. Every team.  
*The same answer*, and a  
test that proves it tomorrow.

---

LIVE DEMO

[sc-ontology-  
app.vercel.app](https://sc-ontology-app.vercel.app)

TEAM

3M-ONTOLOGIST · Mahantesh,  
Devansh, Sathyam

NOTE

The broken definition is left deployed so you can  
watch the test catch it.